

The Treasure Overlord

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

There are $N \times M$ treasure chests arranged in a grid. Each cell contains exactly one chest with a certain amount of **gold**.

Only **one chest** can become the **Treasure Overlord**.

A chest can interact only with its **8 neighboring cells** (up, down, left, right, and the 4 diagonals).

Chests on the border of the grid have fewer than 8 neighbors. Such chests **cannot dominate** and therefore **can never become the Treasure Overlord**.

Domination Rules

A chest at position (i, j) can dominate **all of its 8 neighbors at once** if **at least one** of the following conditions is satisfied:

1. Its gold amount is **strictly greater than the sum** of the gold in all 8 neighboring chests.
2. It is stronger than the **majority** of its neighbors, meaning its gold amount is **strictly greater than at least 5 of its 8 neighbors**.

If neither condition holds, the chest cannot dominate its neighbors.

Determining the Treasure Overlord

Multiple chests may be able to dominate their neighbors.

Among all such chests:

- Compute the **sum of gold values of its 8 neighbors**.
- The **Treasure Overlord** is the chest whose neighbors' sum is the **largest**.
- If multiple chests are tied, choose the one with the **smallest row index**.
- If still tied, choose the one with the **smallest column index**.
- If no chest can dominate its neighbors, print **"PEACE"**.

Input

- The first line contains two integers N and M ($1 \leq N, M \leq 100$).
- The next N lines each contain M integers representing the gold values for the chests ($0 \leq \text{goldvalue} \leq 10$).

Output

Print two integers: the **row index** and **column index** (0-based) of the treasure overlord or "PEACE" according to the rules above.

Examples

standard input	standard output
3 3 2 2 2 2 1 2 2 2 2	PEACE
3 3 1 1 1 1 2 1 1 1 1	1 1
3 3 2 2 2 2 10 2 2 2 2	1 1